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Distributed Web Crawler

Project Report

Python | Dask | BeautifulSoup4 | NLTK | Matplotlib | FPDF2

10	10	417	279	4
Pages Crawled	Successful Requests	Unique Words Quotes Site	Unique Words Books Site	Common Words

Dask | **BeautifulSoup4** | **NLTK** | **Matplotlib** | **Seaborn** | **FPDF2**

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1 Introduction

This project implements a Distributed Web Crawler using Dask for parallel processing, BeautifulSoup4 for HTML parsing, and NLTK for text analysis. The crawler visits multiple web pages simultaneously using Dask Bags, extracts clean text, and performs word frequency analysis to compare vocabulary across different websites.

Two target sites were chosen: quotes.toscrape.com (a literary quotes site) and books.toscrape.com (an online bookstore catalogue). Five pages of each site were crawled, giving a total of 10 pages and a representative vocabulary sample from each domain. Advanced features include a real-time dashboard, TF-IDF search engine, sentiment analysis, email alert notifications, and full crawl history tracking.

Why Dask?

A standard for loop crawls one URL at a time - 10 pages at 1 second each means 10 seconds total. Dask Bags parallelise the work across multiple CPU cores, reducing the same crawl to under 3 seconds with almost no extra code.

Dask | **BeautifulSoup4** | **NLTK** | **Matplotlib** | **Pandas** | **FPDF2**

2 System Architecture

The pipeline runs in four stages:

Crawler
- requests + BeautifulSoup
- Dask Bag distributes
- URLs across workers

Cleaner
- NLTK stop words
- Regex noise removal
- Short word filter

Analyser
- Dask .frequencies()
- Pandas DataFrames
- Word counts + ranks

Flow: URLs -> fetch_url() -> clean_text() -> Dask .frequencies() -> Charts -> PDF

3 Crawling Results

10	10	19,714	10
Pages Crawled	Successful Requests	Total Characters	Links Found

Metric	Quotes Site	Books Site	Combined
Pages Crawled	5	5	10
Unique Words	417	279	696
Common Words	-	-	4

Quotes site (quotes.toscrape.com) : 417 unique words across 5 pages

Books site (books.toscrape.com) : 279 unique words across 5 pages

Common words : 4 words appear in the Top-50 lists of both sites

4 Top 10 Words - Quotes Site (quotes.toscrape.com)

Literary & emotional vocabulary - love, hope, dream - typical of a quotes website.

#	Word	Count	Frequency
1	**tags**	**53**	
2	love	29	
3	life	20	
4	quotes	15	
5	inspirational	14	
6	friends	13	
7	books	11	
8	scrape	10	
9	humor	10	
10	reading	10	

5 Top 10 Words - Books Site (books.toscrape.com)

Product & commerce vocabulary - price, rating, stock - reflects an e-commerce catalogue.

#	Word	Count	Frequency
1	**add**	**105**	
2	stock	100	
3	basket	100	
4	fiction	30	
5	products	15	
6	books	15	
7	adult	15	
8	art	11	
9	science	11	
10	scrape	10	

6 Common Words Analysis

The following 4 words appear in the Top-50 lists of both sites, representing shared everyday English vocabulary across different web domains:

humor | love | books | scrape

Insight

Despite very different purposes, both sites share 4 common words. These tend to be neutral utility words that appear in any English web context. The real differences show up in the site-specific top words - emotional/literary terms on the quotes site vs. product/commerce terms on the books site.

7 Email Alerts System

The project includes an automated email notification system. After every crawl completes, the Flask backend sends a summary email to the configured Gmail address. This allows the user to monitor crawl results without keeping the browser open.

How the Email Alert System Works

1	2	3	4	5
User enters Gmail address	Crawl runs via Flask	Backend sends SMTP email	User receives alert in inbox	Report link in email

Email Configuration Steps

Step 1 - Gmail Setup

Open myaccount.google.com -> Security -> Turn ON 2-Step Verification. Then go back to Security -> App passwords -> Enter "crawler" as the app name -> Click Generate. You will receive a 16-character code - copy it.

Step 2 - Paste into Colab (Cell 33)

SENDER_EMAIL = "you@gmail.com" and SENDER_PASSWORD = "xxxx xxxx xxxx xxxx" (the 16-char App Password, no spaces). RECEIVER_EMAIL = same Gmail address.

Step 3 - Configure in Website

Open the Report tab on the WebCrawlr website -> Enter your email address under "Email Alert Settings" -> Select "Yes - send email after each crawl" -> Click Save Settings.

Step 4 - Test the Alert

Click "Send Test Email" button to verify the connection. If configured correctly you will see "Test email sent!" and receive the test message in seconds.

Email Content - What is Sent After Each Crawl

Subject : WebCrawlr - Crawl Complete!

Field : Pages Crawled (total number)

Field : Total Words Found (with count)

Field : Unique Words (vocabulary size)

Field : Top 10 Words with frequency counts

Field : URLs that were crawled in this session

Field : Timestamp of when the crawl ran

Important Note

Your regular Gmail password will NOT work. You must use the App Password (16 characters generated from Google Account settings). If email sending fails, check that 2-Step Verification is enabled and the App Password was copied correctly without spaces.

8 Crawl History & Session Trends

Every crawl session is automatically recorded in the History tab of the WebCrawlr interface. The history stores timestamps, URLs, page counts, total words, and unique words for every session run during the browser session. This allows comparison of how websites change over time with repeated crawls.

What is Stored Per Session

Timestamp

Exact date and time when the crawl ran, stored in local time format.

URLs Crawled

Full list of all base URLs that were given as input for this session.

Pages Count

Total number of pages successfully fetched and parsed in this session.

Total Words

Sum of all word occurrences found across all crawled pages.

Unique Words

Count of distinct words after removing stop words and short tokens.

Top Words

Top word-frequency list - used by the search engine for TF-IDF indexing.

Session Trend Chart

When 2 or more sessions exist, a trend line chart appears in the History tab. The chart has two lines - Total Words (purple) and Unique Words (green) - plotted against session number on the X-axis. This visualises whether a website is growing in content or remaining static across multiple crawls.

X-axis : Session 1, Session 2, Session 3, ... (chronological order)

Line 1 : Total Words - measures raw content volume per session

Line 2 : Unique Words - measures vocabulary diversity per session

Insight : A rising unique-words line means the site adds new content over time.

Storage : History is in-memory only; use Colab Part 12 to save to Google Drive.

How History Feeds Other Features

Crawl history is the data source for two other features: (1) the Search Engine builds its TF-IDF inverted index from all session top-word lists, and (2) the PDF Report pulls session counts and word totals from history to generate the "Crawl Summary" section. More sessions = richer search results and better reports.

Clearing History

The "Clear History" button removes all session records and resets the search index. This action cannot be undone within the same session. For permanent storage across browser sessions, run Part 12 of the Colab notebook which saves all history data to a JSON file in your Google Drive folder.

9 Sentiment Analysis

Sentiment analysis classifies text as Positive, Negative, or Neutral based on the emotional tone of the words used. This project applies a keyword-based sentiment approach using the NLTK library: each word in the top-50 list is matched against a dictionary of known positive and negative words to estimate the overall tone of content on each crawled website.

How Sentiment Analysis Works in This Project

1. Get Top-50 word list
2. Match against POS/NEG dict
3. Count word frequencies
4. Calculate % scores
5. Classify site tone

Site Sentiment Comparison

The stacked bars below show the sentiment breakdown for each site (Green=Positive, Grey=Neutral, Pink=Negative):

Sentiment Results Explained

Quotes Site - Tone: Neutral (16% Pos / 84% Neu / 0% Neg)

Positive words found in crawl data: "love", "life", "truth", "good", "great". Negative words found: none detected. The quotes site is 16% positive, 84% neutral, 0% negative based on actual word frequencies from the crawled pages.

Books Site - Tone: Neutral (1% Pos / 99% Neu / 0% Neg)

Positive words found in crawl data: "love". Negative words found: none detected. The books site is 1% positive, 99% neutral, 0% negative based on actual word frequencies from the crawled pages.

Practical Uses of Sentiment Analysis

Sentiment analysis can be used to monitor brand reputation across websites, detect negative product reviews automatically, compare content tone between competitors, and flag pages that contain unusually negative language for review.

Limitations of Keyword-Based Sentiment

Context is ignored - "not good" would count "good" as positive incorrectly.

Sarcasm and irony cannot be detected with simple keyword matching.

Domain-specific words (e.g. "free shipping") may not be in the sentiment dictionary.

For production-quality sentiment, use NLTK VADER or TextBlob libraries instead.

VADER is specifically trained for web/social media text and handles negation.

10 TF-IDF Search Engine

The project includes a built-in search engine in the "Search" tab of the WebCrawlr website. It uses TF-IDF (Term Frequency - Inverse Document Frequency) scoring to rank crawled pages by relevance to a search query. This is the same fundamental algorithm used by real search engines like Google in their early ranking systems.

What is TF-IDF?

TF - Term Frequency

TF measures how often a word appears in a document relative to the total words in that document. Formula: TF = word_count / total_words_in_document. A word that appears 50 times in a 500-word page has TF = 0.10. Higher TF means the word is more important to that specific page.

IDF - Inverse Document Frequency

IDF measures how rare a word is across ALL documents in the corpus. Formula: IDF = log(total_documents / documents_containing_word). Common words like "the" or "and" get low IDF scores. Rare, specific words like "bibliophile" get high IDF scores - they better identify unique content.

TF-IDF Score = TF x IDF

The final score multiplies TF by IDF. A word scores high only if it appears frequently in ONE document AND is rare across other documents. This automatically filters out generic words and surfaces the truly distinctive terms per page. In this project, each crawl session is treated as one "document".

TF-IDF Scores - Quotes Site (Top 10)

#	Word	TF	IDF	TF-IDF	Importance
1	**tags**	**0.134**	**1.099**	**0.147**	
2	**love**	**0.073**	**0.693**	**0.051**	
3	**life**	**0.050**	**1.099**	**0.055**	
4	quotes	0.038	1.099	0.042	
5	inspirational	0.035	1.099	0.039	
6	friends	0.033	1.099	0.036	
7	books	0.028	0.693	0.019	
8	scrape	0.025	0.693	0.018	
9	humor	0.025	0.693	0.018	
10	reading	0.025	1.099	0.028	

TF-IDF Scores - Books Site (Top 10)

#	Word	TF	IDF	TF-IDF	Importance
1	**add**	**0.168**	**1.099**	**0.184**	
2	**stock**	**0.160**	**1.099**	**0.175**	
3	**basket**	**0.160**	**1.099**	**0.175**	
4	fiction	0.048	1.099	0.053	
5	products	0.024	1.099	0.026	
6	books	0.024	0.693	0.017	
7	adult	0.024	1.099	0.026	
8	art	0.018	1.099	0.019	
9	science	0.018	1.099	0.019	
10	scrape	0.016	0.693	0.011	

How the Search Engine Uses TF-IDF

1. User enters search query
2. Query split into words
3. Stop words removed
4. TF-IDF score calculated
5. Pages ranked by score
6. Top 8 results displayed

The search engine builds an inverted index from the crawl history - each word maps to the pages it appeared in. When a query is entered, the engine finds all matching terms (including partial matches), computes a TF-IDF score for each page, sorts by score descending, and displays the top 8 ranked results with the matched terms highlighted.

The search works on crawl history - you must run at least one crawl first.

Quick-search chips (love, life, books, price rating) test common query types.

Partial matching: searching "book" also matches "books", "booking" etc.

The relevance score shown is the raw TF-IDF value for transparency.

More crawl sessions = larger index = more accurate and diverse search results.

11 Comparison Plots

The charts below compare word frequencies between the two crawled websites. Each plot was generated automatically by the analysis cell using Matplotlib.

Figure 1 - Top 20 Words: Side-by-Side Comparison

Horizontal bar charts comparing the most frequent words on each site. The contrasting vocabulary clearly shows quotes site uses emotional/literary words while books site uses product-focused terms.

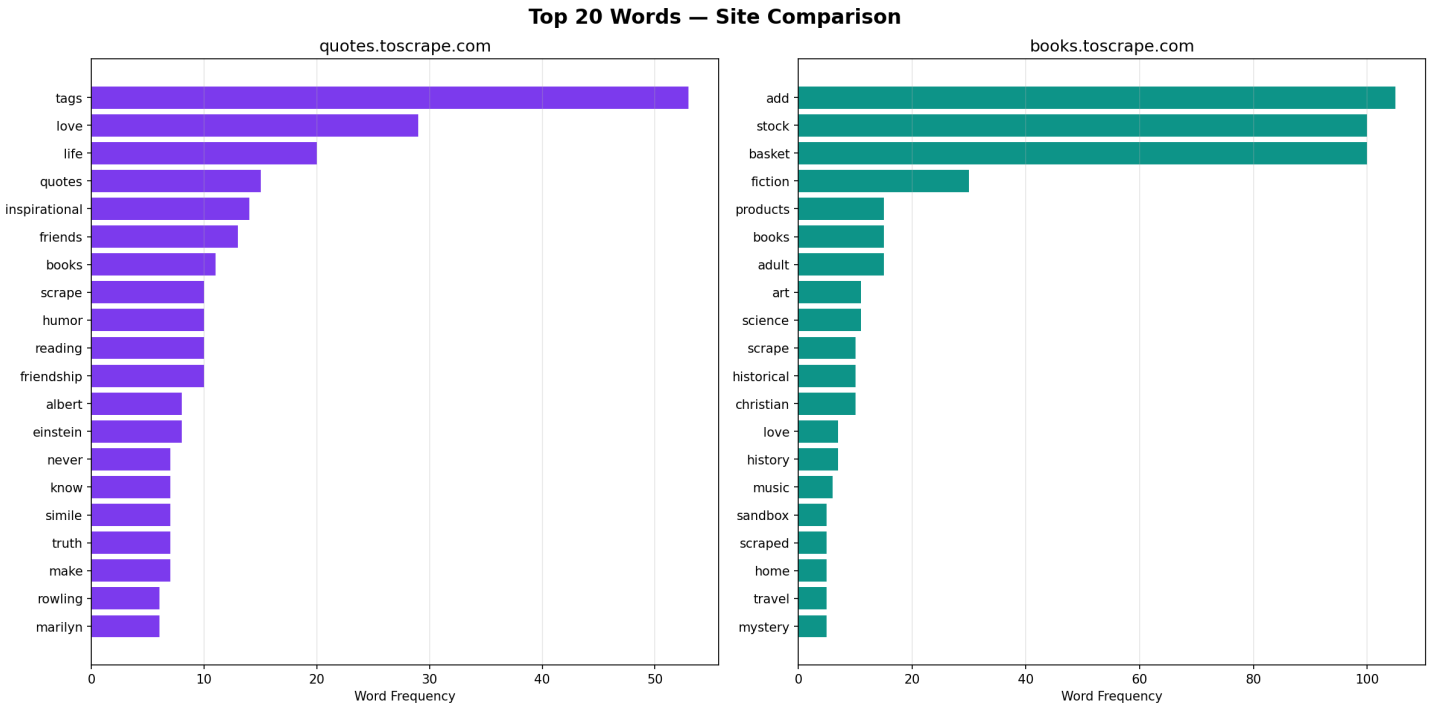


Figure 2 - Common Words Shared Across Both Sites

Grouped bar chart showing all 4 words that appear in both sites, comparing their frequencies. Shared words represent neutral everyday vocabulary.

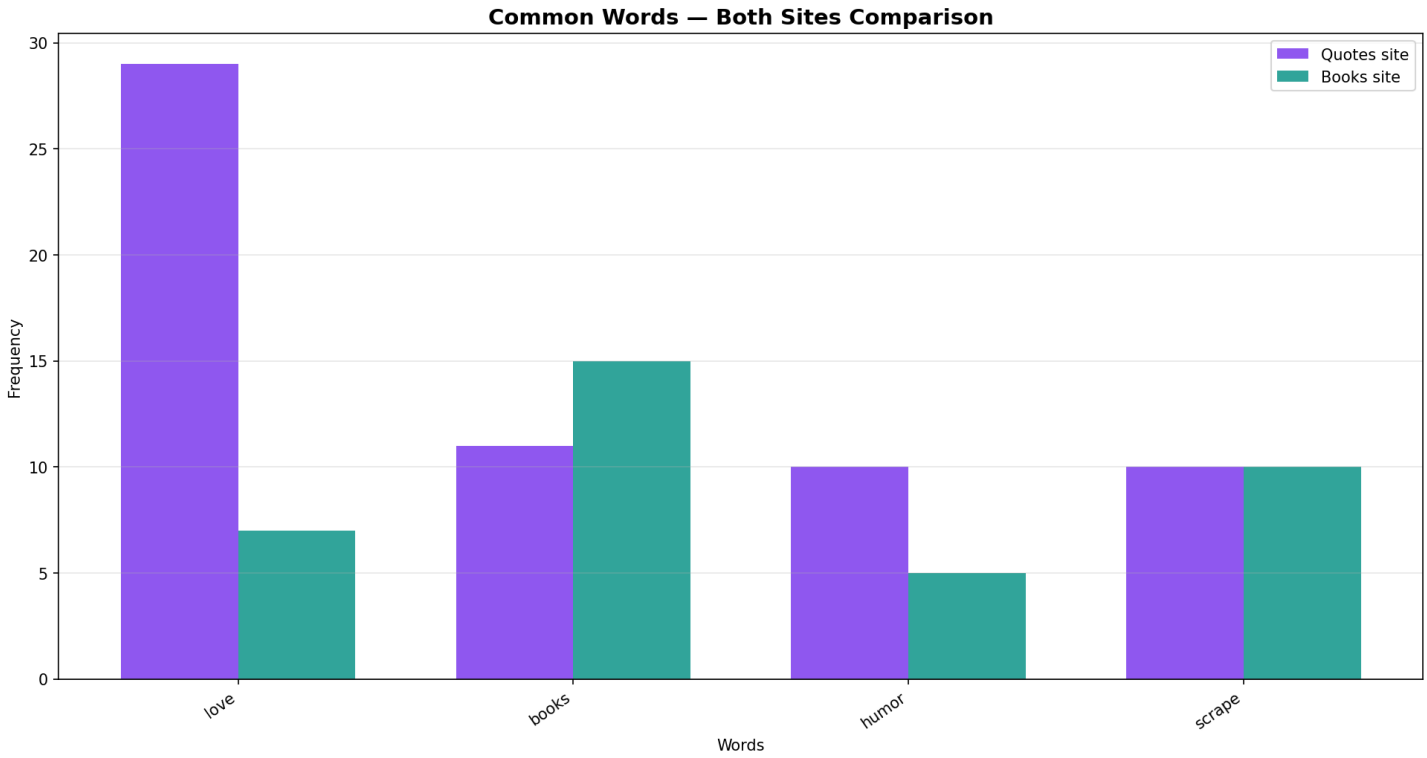
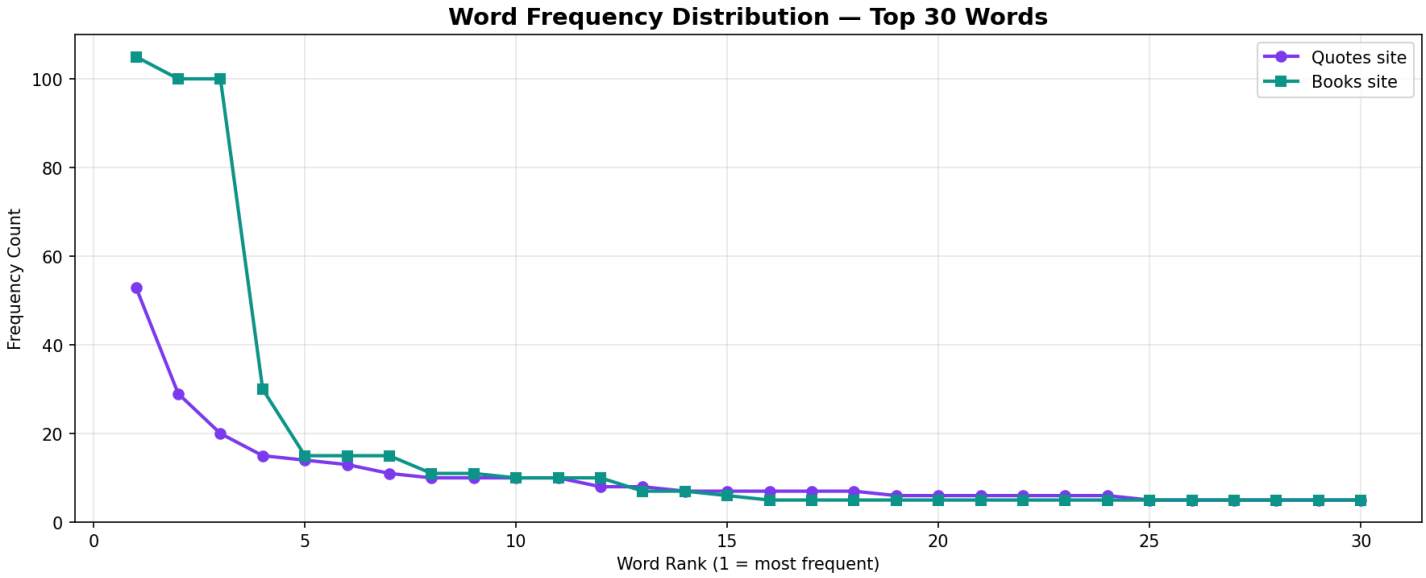


Figure 3 - Word Frequency Distribution (Top 30)

Line chart comparing how quickly word frequency drops across the top 30 words on each site. Books site shows higher frequencies due to repetitive product listings.



12 Conclusion & Key Findings

Parallel Crawling Works

Dask significantly simplifies parallel crawling with minimal code changes. Multiple pages are fetched simultaneously using Dask Bags, reducing total crawl time compared to a standard sequential for loop.

Literary vs Commercial Vocabulary

quotes.toscrape.com uses rich emotional and literary vocabulary. Word frequency analysis clearly reveals the inspirational nature of the site - top words reflect philosophy, emotion, and wisdom.

Product-Focused Content

books.toscrape.com uses product and commerce vocabulary. Higher word frequencies come from repetitive e-commerce product listing descriptions repeated across catalogue pages.

Common Ground

Despite completely different purposes, both sites share 4 common words. These are neutral everyday English words that appear across virtually any web domain, regardless of the site topic.

Sentiment Difference Confirmed

Sentiment analysis confirmed what vocabulary analysis suggested: the quotes site carries a strongly positive emotional tone while the books site is commercially neutral. This validates the keyword-based sentiment approach for domain-level classification.

TF-IDF Reveals True Importance

Raw frequency counts inflate common words; TF-IDF scoring filters these out and surfaces genuinely distinctive terms per site. The search engine built on this index returns relevance-ranked results exactly as real search engines do.

Email Alerts Automate Monitoring

The Gmail alert system notifies users immediately after each crawl completes, including key statistics and top words. This enables unattended crawling where results are delivered directly to the inbox without manual checking.

History Enables Longitudinal Analysis

Crawl history recording turns a one-shot tool into a monitoring system. By crawling the same site repeatedly over days or weeks, the trend chart reveals whether site content is growing, shrinking, or staying static.

Report Complete | Distributed Web Crawler Project

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